

Sachin Kumar Sharma

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SUMMARY

Data Engineer with 6 years architecting distributed data platforms on GCP and AWS — spanning Hadoop-to-cloud migrations, PySpark pipeline optimisation, and Python backend microservices/APIs. Delivered **50% faster pipeline execution** and **40% compute cost reduction**, and architected an **MCP-based Generative BI platform** integrating LLMs with BigQuery for natural language querying. GCP-certified; Fortune 500 experience in financial services.

EXPERIENCE

American Express Apr 2025 – Present

Data Engineer III • Bengaluru, India

- Architected end-to-end migration of large-scale on-premise pipelines to **cloud-native infrastructure**, eliminating single-point hardware dependencies and enabling elastic, on-demand scaling.
- Redesigned core data architecture around decoupled storage/compute layers, reducing deployment complexity by **~35%** and establishing a more maintainable, extensible platform foundation.
- Engineered and tuned PySpark pipelines using partition pruning, broadcast joins, and dynamic resource allocation, achieving **50% faster execution and 40% lower compute costs**.
- Directed orchestration of multi-stage Airflow workflows across cloud object storage and columnar warehouses; established structured logging and audit tables for **real-time SLA tracking**.
- Designed and built a semantic layer on BigQuery to standardise business metrics, then architected an **MCP-based Generative BI platform** integrating LLMs with BigQuery for natural language querying and automated chart generation, eliminating manual SQL hand-offs.
- Built and maintained **Python backend microservices and RESTful APIs** to expose data pipeline outputs and enable seamless, scalable integration with downstream business applications.

Tech Stack: Apache Spark, PySpark, Airflow, GCP (BigQuery, Dataproc, Dataflow, Cloud Storage, Pub/Sub), Python, SQL, MCP, LLM Tool Calling, Natural Language to SQL, Generative BI, Python Microservices, REST API Development

Deloitte South Asia Jun 2023 – Apr 2025

Consultant – Data Engineering • Bengaluru, India (Onsite: Dubai, UAE — Oct–Dec 2023)

- Led end-to-end pipeline delivery for enterprise clients using Hadoop, Hive, PySpark, Airflow, and Control-M, supporting **petabyte-scale** processing requirements.
- Designed star/snowflake warehouse schemas and optimised partition strategies, measurably reducing query latency across client platforms.
- Architected scalable cloud pipelines on BigQuery and GCS/S3; established automated data-quality and anomaly-detection frameworks, proactively reducing critical data incidents.
- Dubai engagement: designed a unified customer data model and real-time alerting framework for a **Tier-1 bank's** financial crime analytics function, **cutting false-positive investigation workload**.

Tech Stack: Hadoop, Hive, PySpark, Airflow, Control-M, BigQuery, GCS, AWS S3, Python, SQL

Wipro Technologies Sep 2020 – Jun 2023

Senior Data Engineer • Bengaluru, India

- Engineered complex ETL pipelines (HDFS, Hive, Spark, SQL, Control-M) to process high-volume data from SQL Server, Oracle, and Salesforce into centralised warehouses.
- Designed full-load, incremental, and CDC ingestion patterns, improving data freshness and reducing ingestion windows.
- Led the on-premise-to-cloud migration, replatforming Hadoop workloads to managed Spark and cloud warehousing with **zero data loss**; applied caching, partitioning, and shuffle-reduction tuning for measurable throughput gains.

Tech Stack: HDFS, Hive, Apache Spark, PySpark, SQL, Control-M, Python, SQL Server, Oracle, Salesforce

SKILLS

Languages: Python, SQL, HiveQL

Big Data: Apache Hadoop, Apache Hive, Apache Spark, PySpark, HDFS, Databricks

Cloud: GCP (BigQuery, Dataproc, Dataflow, Cloud Storage, Pub/Sub), AWS S3

Databases: BigQuery, MySQL, PostgreSQL, SQL Server, Teradata, Nessie, DremIO, MinIO

Backend & APIs: Python Microservices, RESTful API Development, Microservices Architecture

ETL / Tools: Airbyte, Apache Airflow, Control-M

DevOps: Git, CI/CD pipelines, Elasticsearch, Postman, VS Code

AI / LLM Integration: MCP (Model Context Protocol), LLM Tool Calling, Natural Language to SQL, Generative BI

Data Modelling: Star Schema, Snowflake Schema, Dimensional Modelling, CDC, Data Vault

CERTIFICATIONS

- Google Cloud Certified Professional Data Engineer
- Google Cloud Certified Associate Cloud Engineer

EDUCATION

Vellore Institute of Technology (VIT), Chennai Jul 2016 – Jul 2020

B.Tech in Electronics and Communication Engineering • CGPA: 8.43 / 10